User Guide: FEM-Long Overhung Rotor

Introduction & Underlying Physics

These MATLAB programs were a group of modular codes that work together to build a finite element model, dedicated for long overhung rotor, and do simulations of the rotor in operating condition while intermittent contact occurs between rotor and stator. While the actual systems (e.g. sump pump, drill string, turbine) were possibly running through contact on multiple locations, currently the programs are limited to maximum 2 contact points.

The system motion is governed by discretized finite element equations incorporating speed-dependent gyroscopic coupling and non-linear rotor-stator contact forces:

$$\boldsymbol{M}\ddot{q}(t) + [\boldsymbol{C} + \Omega\boldsymbol{G}]\dot{q}(t) + \boldsymbol{K}q(t) = \boldsymbol{F_{unbalance}}(t) + \boldsymbol{F_{contact}}(t)$$

From the equation, $\boldsymbol{M}$, $\boldsymbol{C}$, $\boldsymbol{G}$, and $\boldsymbol{K}$ represent global mass, damping, gyroscopic, and stiffness matrices, Ω represents rotational speed, and $q(t)$ represent rotor displacement and angular response over time.

The nonlinearity of contact force lies in the function $\boldsymbol{F_{contact}}(t)$ where the value only exists when contact occurs. This force is a function of normal and tangential force:

- Normal force, a function of contact stiffness and damping
$$F_{normal} = k_{contact} \times (|r - c|) + c_{contact} \times \dot{r}$$
- Tangential force, a function of coefficient of friction and normal force itself
$$F_{tangential} = \mu \times F_{normal}$$

Physically, the normal force only appears when the contact occurs, on the contrary the value will be zero. The force due to contact stiffness determined by how much penetration, radial displacement (r) subtracted by radial clearance (c), occurs. Meanwhile, force due to contact damping as a function of radial speed ($\dot{r}$) might negate the normal force when the velocity pulling the rotor away from stator. In the actual condition, it was unlikely to happen, so adjustment was made to prevent the force value to be zero.

The current model was an overhung rotor with 2 supports and 2 contact points (maximum). This model can be used to simulate system's displacement, velocity, and acceleration of each node from the model. In addition, contact forces in both axes were obtainable for analysis. This package also provide basic analysis tools such as FFT, orbit, and overall orbital response.

Program overview

To use the FE model, simply download all the program and put them all in one folder. These programs mainly have 2 functions, to do the simulation and to interpret the results. The first function is handled by several codes run in parallel:

- **Main_Program.m** → This is what you run to do the simulation.
- **Parameter.m** → This is where you input systems configuration (model characteristics, element distribution, and simulation configuration).
- **Matrix_Prep.m** → This is the matrix builder for the model, to adjust system's matrix based on element distribution defined by user.
- **Run_up.m** → This is the Newmark-β solver for time response integration. Except you want to change the coefficients related to this integration method, preferably to not open/change anything from this program.

The second function is handled by other programs:

- **Result_Compiler.m** → This is what you run to analyze the simulation results to see the desired results.
- **Res_FFT.m** → Shows FFT result of vibration on a certain node.
- **Res_Orbit.m** → Shows the orbit of a certain node.
- **Res_OBOVR.m** → Shows the orbit and deflectional shape of all nodes (OBOVR stands for Orbit-Overall).
- **Res_OVT.m** → Shows the displacement, velocity, acceleration, and force given to system overtime.
- **Campbell** → Shows Campbell Diagram.

Simulation

To do the simulation, simply open **Main_Program.m** and **Parameter.m** in MATLAB. You can input the rotor, element, and simulation parameters in **Parameter.m**. Important thing to note here is that variable "te" in **Parameter.m** was simulation time – how long the rotor will be operated. While response of the rotor usually analyzed in steady-state, the response at several seconds at the end of simulation were the one being recorded for analysis. This recording time defined by "Tr" in **Main_Program.m**. To reduce the size of simulation result file, increase the value of "dec" in **Main_Program.m** to increase the sampling period (sampling period is the multiplication of "dec" and time step.

The default result will be a .mat file with a name that is self-defined (example: "NoBushing_RPM_2000.mat") , containing 4 matrices:

- t_ss → time
- q_nb_ss → displacement, velocity, and acceleration
- FB1_ss → Force at first bushing/contact point
- FB2_ss → Force at second bushing/contact point

Please note that you can add other matrices if you want to.

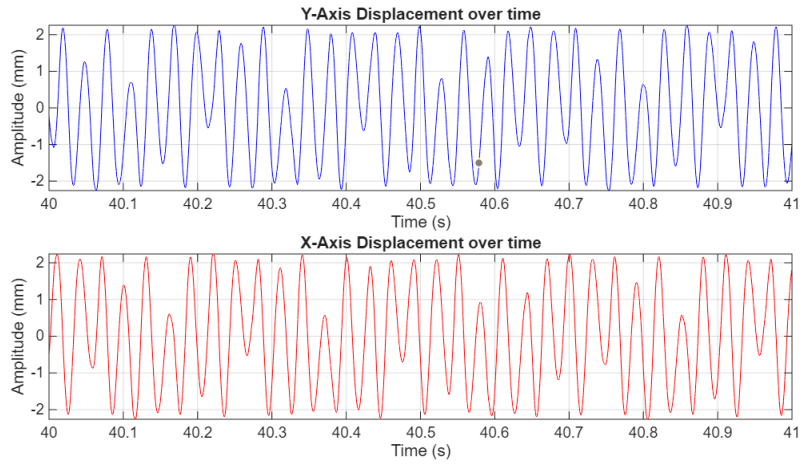
Results interpretation

The result of simulation saved inside the .mat file is in the form of matrices, which is hard to interpret. It was either you want to visualize it by yourself or use several functions that have been defined in these programs. Using **Result_Compiler.m**, you can define certain node that you want to observe by changing variable “n”, specific time of observation by defining “t_Start” and “t_Stop”, and execute some functions.

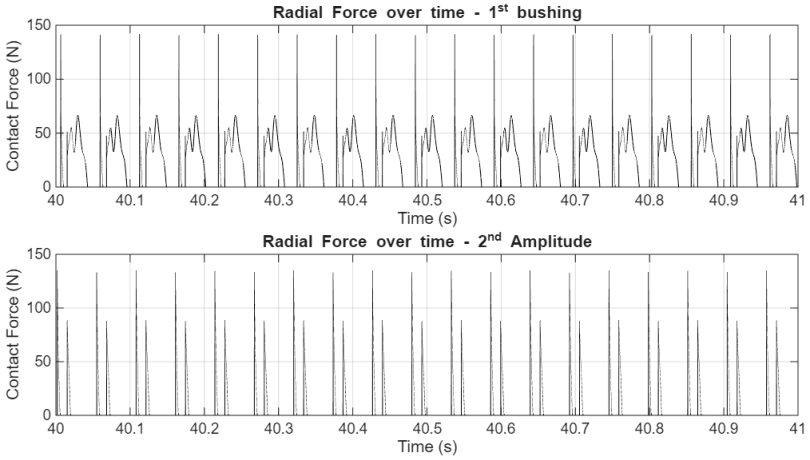
Res_OVT

This function will show you several responses as a function of time.

- Displacement, velocity, and acceleration will only show defined nodes at defined range of time

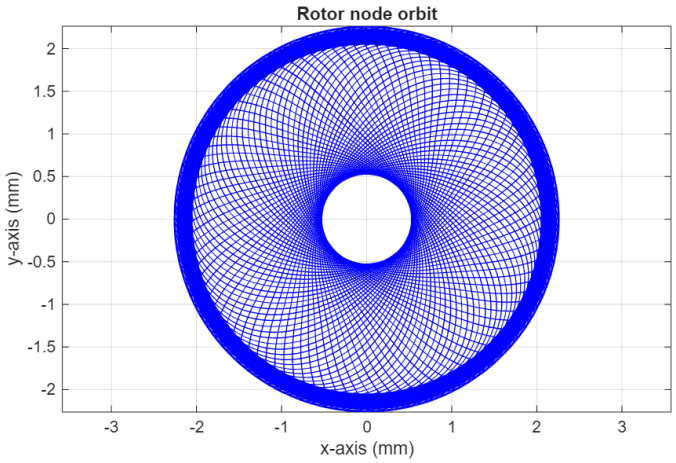


- Contact force will only show for defined range of time for both bushings/contact points



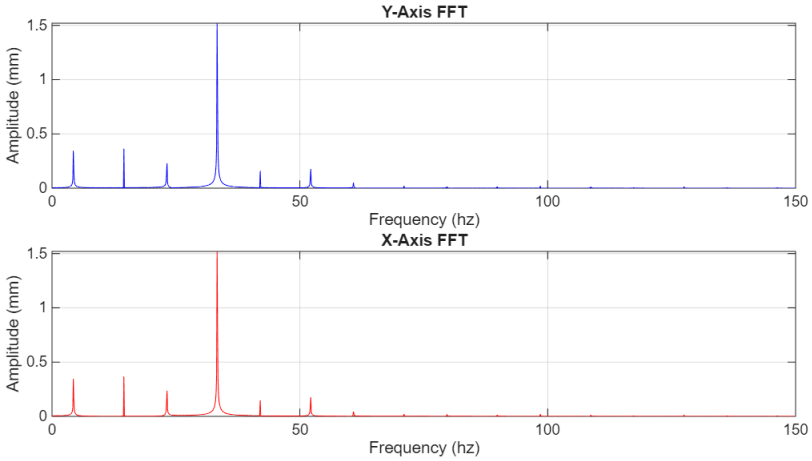
Res_Orbit

This function will show you the radial movement of a defined node at defined range of time



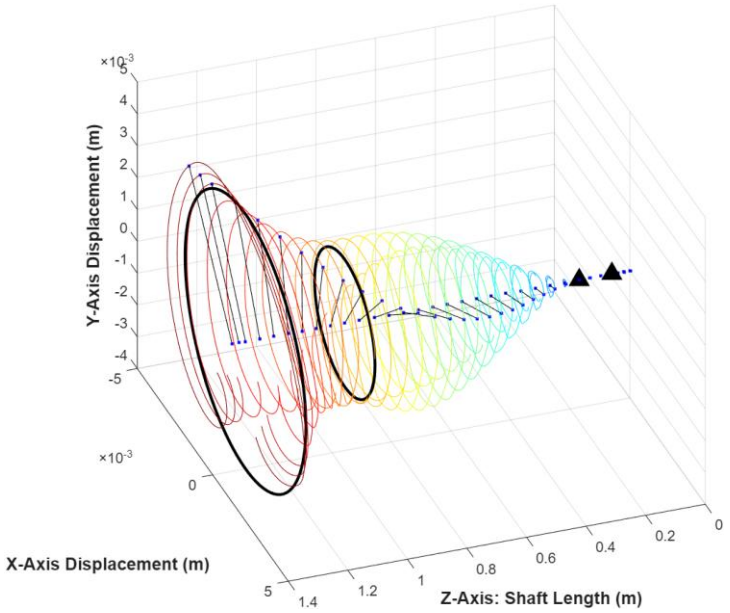
Res_FFT

This function will show you the spectral components of a defined node for entire range of time in the data



Res_OBOVR

This function will show you the orbit of each node of the rotor in a defined range of time. This graphy will show you how the rotor move and deflects, especially when contact occurs



Campbell

This function will show you a Campbell Diagram, which is useful to define the critical speed of the rotor and the mode shape of the rotor for each critical speed.

